Vector Calculus and Complex Analysis Classwork (2025/4/17)

Name:

Student ID:

Consider the function $f(z) = 2x^2 + 3xy + 4y^2 + (5x + y)i,\ z = x + iy.$ Find the region where

(a) the Cauchy–Riemann relations are satisfied;

(b) the function is differentiable;

(c) the function is analytic.

Calculate $f'(z)$ if f is differentiable.

Solution

(a) $u = 2x^2 + 3xy + 4y^2, \quad v = 5x + y$

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \Rightarrow \quad 4x + 3y = 1, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \Rightarrow \quad 3x + 8y = -5$

f satisfies C-R equations when $z = 1 - i$

(b) All 1-st partial derivatives are continuous. f is differentiable when $z = 1 - i$.

(c) f is nowhere analytic because $z = 1 - i$ is not an open connected set.

$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = 1 + 5i$ at $z = 1 - i$.